

Robin CR

Chennai, Tamil Nadu

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Personal Profile

Senior Software Engineer with strong expertise in modern C++ (C++11/14/17), multi-threading in a Linux-based development. Skilled in developing algorithms, modules and utilities for an embedded or application project. Interested in Perception, Path planning, Sensor fusion and SLAM algorithms.

Work Experience

Automotive Robotics India (ARi)

Chennai, Tamil Nadu

Senior Software Engineer

January 2025 – Present

CATERPILLAR (Onsite)

LIDAR Perception for Autonomous Surface Hauling truck

- Engineered Moving Obstacle Detection solutions leveraging lidar sensor.
- Built and optimized range image-to-point cloud converters, enabling support for multiple lidar types.
- Developed rut and berm detection feature for mine site.
- Supported the development of mapping the boundaries of mine site.
- Integrated open-source Point Cloud libraries into C++ projects to accelerate development.
- Drove KPI improvements by investigating and resolving root causes of KPI mismatches.
- Configured and managed ROS TF tree, defining coordinate frames and static transforms via YAML.
- Contributed to the robust backend systems for a playback tool.
- Pioneered a utility for GPU benchmarking to analyze the performance of developed algorithms.
- Developed a utility to visualize vehicle movement with the help of plotly library.
- Performed core dump analysis to identify root causes of system crashes and runtime failures.
- Profiling, call graph and memory footprint analysis for the developed modules.

Bosch Global Software Technologies

Coimbatore, Tamil Nadu

Embedded Software Engineer

August 2022 – December 2024

Interior Monitoring System (IMS) – Camera Perception - Stellantis/Ford

- Developed algorithms for driver drowsiness module in C++.
- Contributed to new features for playback tool for simulating driver behavior.
- Implemented a module to monitor facial features of the driver.
- Developed an algorithm called circadian rhythm to handle drowsiness in drivers.
- Contributed to the development of hardware and software reset for camera.
- Modules implementations such as interfaces, precondition handlers, error handlers, schedulers etc.
- Contributed to the development of monitoring the GPU load for model training.
- Successfully addressed and resolved Level 8 and 9 QAC++ warnings in the codebase.
- Worked on the project migration from virtual machine to docker.

Digital Instrument Cluster - Volvo

- Worked in all phases of Agile Scrum methodology (Requirement - Design - Coding - Testing).
- Contributed to the development and integration of the Trip Computer module.
- Identified and resolved a critical tell-tale toggling issue.
- Investigated, raised, and fixed a bootloader timeout issue in the ECU.
- Worked on the ASPICE-related standards followed in AUTOSAR projects and handled the audit.
- Implemented an algorithm that will modify the fixed parameters in the RTE layer into a configurable parameter in the application layer.
- Base level knowledge of Diagnostics, Bootloader and RTE layer is gained.

Gadgeon Systems Pvt Ltd

Kochi, Kerala

Embedded Software Developer- Gateway

July 2021 - August 2022

- Interfaced accelerometer, gyroscope, and IMU sensors with an ARM-based microcontroller.
- Developed codecs and custom protocols for different sensors in a Real Time Linux environment.
- Integrated the GPS with the cloudgate gateway to monitor the truck locations in the warehouse.
- Active contributor to the development of BSP for a Nordic Semiconductor board.

Projects

- **CPU-GPU Selector** | [Link](#)
Implemented a system that decides the model training to be done on CPU, GPU, or GPU on the cloud.
- **Calendar and Clock** | [Link](#)
Project that handles the tricky scenarios related to time.
- **Smart Trolley System** | [Link](#)
A simple project which plans for the nearest path for the trolley when given an item in a supermarket.

Skills

Microcontrollers	ARM Cortex M3 LPC1768, ESP8266
Lidar	Velodyne, Luminar, Ouster
Protocols	UART, I2C, SPI, CAN
Design	Enterprise Architecture, PlantUML
Debugger	Valgrind, GDB, Lauterbach Trace32
Tools	CANoe, Perf
Libraries	Gtest, BOOST, EIGEN, PCL, STL
Version control system	Git, RTC, BitBucket
Build	Cmake, Sconsript
AI tools	Github Copilot, Cursor

Competencies

- **Embedded C programming**
State machine, Bit wise programming, handshaking mechanisms.
- **C++ Application development**
Move Semantics, Smart Pointers, Templates, Functional Programming, Atomics.
- **AI Integration**
Synthetic dataset generation using NVIDIA Cosmos, Local LLM integration, Agentic AI

Certifications

Embedded System Professional

Translot, Kochi

- Solid foundation on the basics of embedded C programming.
- Completed the projects based on EEPROM and display systems.
- Working knowledge of POSIX programming is achieved.
- Proper understanding of Linux Fundamentals (System calls, Booting, file system) is gained.

Professional C++ and Clean code

Bosch, Coimbatore

- Leveraged the OOPs and SOLID principles with examples.
- Discussed common pitfalls in C++ for embedded software development.
- Covered Design Patterns explaining their roles in code reusability and scalability.
- Clean coding guidelines and common practices were addressed.

Agentic AI

Claude, Online

- Applied agentic AI concepts including skills, hooks, and agent architectures.
- Practiced prompt engineering techniques for AI-assisted software development workflows.
- Studied the Model Context Protocol (MCP) and its role in extending AI capabilities.
- Gained knowledge of using Claude for research.

Education

Rajagiri School of Engineering and Technology

Bachelor of Technology in ECE
Completed with a GPA of 8.2

Kochi, Kerala
Aug 2017 - Jul 2021